

A Dynamic DCA-Leverage Algorithm for Optimized Average Positioning and Reduced Liquidation Risk

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1970-01-01

This paper presents a novel dynamic DCA (Dollar Cost Averaging) combined with leverage allocation algorithm designed to optimize average entry pricing and reduce liquidation risk in volatile markets. The algorithm defines DCA points based on a controlled drawdown function:

$$D_k = f + k \frac{((n+1) \cdot 2 - f)}{n}, k = 0, 1, 2, \dots, n$$

where f represents the initial DCA percentage, n the total number of DCA entries, and k the DCA step index. By combining this DCA schedule with an increasing leverage sequence (e.g., 2x, 4x, 6x, 7x, 10x), the approach ensures that larger positions are taken at lower prices, improving average entry and overall risk management. Simulated scenarios demonstrate that increasing leverage produces superior average entry prices, maintains a safer liquidation buffer, and effectively scales invested amounts to maximize gains. Comparative analysis shows that decreasing or uniform leverage sequences result in higher average entry prices and higher liquidation risk. This research provides a systematic framework for traders seeking to combine DCA with leverage efficiently.

Introduction

Dollar Cost Averaging (DCA) is a common strategy in which an investor allocates a fixed amount of capital at regular price intervals, mitigating the risk of buying at a peak. Leverage amplifies position size but increases liquidation risk.

Combining DCA with leverage allows traders to control risk while optimizing capital efficiency. A key challenge is to determine:

1. Where to place each DCA entry
2. How to scale leverage per entry to balance profit potential and liquidation safety

The proposed algorithm addresses these challenges by creating a dynamic DCA schedule and matching leverage strategy that improves average entry, reduces liquidation probability, and scales invested amounts efficiently.

Mathematical Model

DCA Formula

The DCA positions are determined using:

$$D_k = f + k \frac{((n+1) \cdot 2 - f)}{n}, k=0, 1, 2, \dots, n$$

where:

- f = initial DCA entry in % drawdown
- n = number of DCA steps
- k = current DCA step

Example: $f=1.5\%$, $n=4$

k	Drawdown % (D_k)
0	1.5
1	3.625
2	5.75
3	7.875
4	10.0

This evenly spaces the DCA entries from the first buy point f to the target maximum drawdown.

Leverage Allocation

We define a leverage sequence $L = \{L_0, L_1, \dots, L_n\}$. The proposed increasing leverage approach:

$$L = 2, 4, 6, 7, 10$$

Weighted Average Entry Price:

$$\text{Average Entry} = \frac{\sum_{i=0}^n P_i \cdot L_i}{\sum_{i=0}^n L_i}$$

- Larger leverage at lower prices reduces average entry price and increases the liquidation buffer.
- Decreasing leverage results in higher average entry and higher liquidation risk.

Scaling Investment Amounts

If a base amount A is invested per unit leverage:

$$\text{Effective Investment}_i = A \cdot L_i$$

Example: - 1st DCA: 2x leverage, invest \$5 → effective \$10 - 2nd DCA: 4x leverage, invest \$5 → effective \$20

This ensures larger positions at lower prices, maximizing returns if the market rebounds.

Comparative Analysis

Average Entry

Assume DCA prices: 100 → 95 → 90 → 85 → 80

Increasing Leverage 2,4,6,7,10:

$$\text{Avg Entry} = \frac{100 \cdot 2 + 95 \cdot 4 + 90 \cdot 6 + 85 \cdot 7 + 80 \cdot 10}{2 + 4 + 6 + 7 + 10} \approx 86.7$$

Decreasing Leverage 10,8,6,4,2:

$$\text{Avg Entry} = \frac{100 \cdot 10 + 95 \cdot 8 + 90 \cdot 6 + 85 \cdot 4 + 80 \cdot 2}{10 + 8 + 6 + 4 + 2} \approx 93.3$$

Overall Applied Leverage

$$\text{Avg Leverage} = \frac{\sum_i L_i}{n+1}$$

- Increasing leverage: $(2+4+6+7+10)/5 = 5.8x$ - Decreasing leverage: $(10+8+6+4+2)/5 = 6.0x$

Even with similar average leverage, increasing leverage allocates more effective exposure at lower prices.

Liquidation Buffer

- Average entry closer to cheaper levels → liquidation distance from current market price is greater. - Decreasing leverage uses most exposure at expensive positions → higher liquidation risk.

Investment Efficiency

- Increasing leverage allocates more capital to cheaper entries: - 2x → \$10, 4x → \$20, ... 10x → \$50 (mock example) - Optimally scales DCA amounts to maximize profit while controlling risk.

Simulated Scenarios (Mock Data)

Scenario	Max Drawdown	Avg Entry	Avg Leverage	Liquidation Buffer	P/L (%)
Increasing leverage	10%	86.7	5.8x	13%	+4.5
Decreasing leverage	10%	93.3	6.0x	8%	+2.1
Uniform leverage	10%	90.0	6.0x	10%	+3.0

Conclusion

The dynamic DCA-leverage algorithm allows traders to:

1. Place DCA orders evenly over a defined drawdown.
2. Apply increasing leverage to maximize effective investment at cheaper prices.
3. Reduce average entry price and improve liquidation buffer.
4. Scale investment amounts efficiently to maximize returns.

Comparative analysis demonstrates that increasing leverage is superior to decreasing or uniform leverage sequences. This framework provides a systematic approach for combining DCA and leverage efficiently in volatile markets.

Code and Data Availability

For readers interested in the detailed calculations, formulas, and sample data used to generate the DCA and leverage plots presented in this paper, the complete dataset and Python scripts are available on GitHub. These resources include the implementation of the DCA formula, leverage allocation logic, and simulated scenarios used to illustrate the algorithm's performance. The materials can be accessed at:

<https://github.com/ayushk2s/Best-Averaging-And-Leveraging>. This repository provides additional context for reproducing the figures and conducting further analysis.